

Phone Dialplan Guide

How a stock AllStarLink 3 node handles phone calls, and what the ASL-DVS Pi Zero 2 W dashboard changes

Covers Dashboard Pi02w 9.3.71.10 and SysMon Pi02w 6.13.67.4 (2026-10-06). The full build (Pi 3, 4 or 5) has the same Phone tab and dialplan from Dashboard 9.3.73 and SysMon 6.13.69, so everything here applies to it too. Stock examples come from the AllStarLink app_rpt project's own sample `rpt.conf` and `extensions.conf`.

What this guide is for

A **dialplan** is the set of instructions Asterisk (the phone engine under AllStarLink) follows for every call: where to send it, what to check first and when to hang up. This guide walks through the stock dialplan that ships with ASL3, then the one the dashboard writes for its Phone tab, and why each change was made. You don't need to be a programmer; each piece is explained in plain words before any config is shown.

The words you need

Word	What it means here
Dialplan	The call instructions, kept in <code>/etc/asterisk/extensions.conf</code> and files it includes.
Context	A named group of instructions, written in square brackets, for example <code>[radio]</code> . A call is always “in” one context.
Extension / pattern	What was dialed. A pattern matches many numbers: <code>_X.</code> means “one digit, then anything”; <code>_NXXNXXXXXX</code> is a 10-digit North American number.
Step (priority)	One instruction for that extension. <code>exten => ...,1</code> , is step 1 and each same <code>=> n</code> , line is the next step.
Goto	Jump to another context. This is how one context hands a call to the next.
Dial	Actually ring the number on a phone network (SIP or IAX2).
rpt()	Connect a call to an AllStar node. <code>rpt(1001,Pv)</code> means: node 1001, phone mode (P), voice keying (v).
Autopatch	app_rpt's built-in phone patch. <code>*61</code> starts it and it collects the digits you key.
Function code	A DTMF code in <code>rpt.conf</code> , such as <code>61 = autopatchup</code> , that runs an action when keyed from the radio.
AstDB	Asterisk's small built-in database. The dashboard uses it for switches the dialplan can read, such as “is the Phone tab open?”
Originate	A command the dashboard sends Asterisk (through its manager connection) to place a call directly.
Duplex / VOX	How the node keys the transmitter. VOX keys only while voice is present; <code>duplex = 0/1</code> is a simplex (one-radio) node.

Part 1 — The stock ASL3 dialplan

A fresh ASL3 install has phone calling turned **off**. Everything needed is present but commented out or pointed at a “no service” message. Two files are involved.

1. rpt.conf: the node and its codes

The node template [node-main] names the dialing context and the function table. The autopatch codes are shipped commented out (the leading ;):

Stock rpt.conf (excerpt)

```
[node-main]()
functions = functions           ; Repeater Function stanza
link_functions = functions      ; Link Function stanza
phone_functions = functions     ; Phone Function stanza
context = radio                ; dialing context for phone

[functions-main]()
99 = cop,6                     ; PTT (phone mode only)
;61 = autopatchup,noct = 1,farenddisconnect = 1,dialtime = 20000 ; Autopatch up
;62 = autopatchdn              ; Autopatch down
```

2. extensions.conf: the call instructions

The stock dialplan has a chain for numbers dialed from the radio. It checks the number against a list of blocked kinds, then hands it to [pstn-out], which by default only plays “no service”:

Stock extensions.conf (excerpt)

```
[radio]
exten => _NXXXXXX,1,Goto(check_route,1${HOMENPA}${EXTEN},1) ; 7 digits: add area code
exten => _1XXXXXXXXXX,1,Goto(check_route,${EXTEN},1)
exten => _X11,1,Goto(check_route,${EXTEN},1)

[check_route] ; refuses 800/888/877/866/855, X00, X11, 555, 976, 809, 900
exten => _1800NXXXXXX,2,Goto(invalidnum,s,1)
...
exten => _1NXXXXXXXXX,2,Goto(pstn-out,${EXTEN:1},1) ; okay, route it

; Comment-out the following clause if you want Allstar Autopatch service
[pstn-out]
exten => _NXXXXXXXXX,1,Wait(1)
same => n,Playback(ss-noservice)
same => n,Hangup
```

Other stock contexts handle node links rather than phones: [radio-secure] (node-to-node connections with allow/deny lists), [iaxprt] and [iax-client] (IAX apps that join a node, the last one with Rpt(node,P,...) phone mode), and [allstar-public]/[allstar-sys].

How a stock autopatch call flows (once it is turned on)

1. You key *61 and the number

The radio node's own function table runs autopatchup.



2. Autopatch collects the digits

It waits for a short pause after the last digit (dialtime).



3. Context radio → check_route → pstn-out

Blocked kinds go to invalidnum; good numbers reach pstn-out.



4. pstn-out dials the phone company

You uncomment pstn-out and point its Dial() at your provider.



5. The call runs as an autopatch on the radio node

*62 (autopatchdn) or the far end hanging up ends it.

Built-in limit: while an autopatch call is up, app_rpt treats the node as keyed toward every linked node for the whole call (the code that would gate this by voice is compiled out because it clipped first syllables). On a full-duplex repeater that's fine. On a simplex hotspot it is a **dead key**: the radio transmits the whole call and can't hear you. Forum reports describe the same thing back to 2009.

Part 2 — The ASL-DVS Pi02w dialplan

The dashboard does **not** edit the stock contexts. It writes its own file and links it in with one line, so the stock dialplan keeps working exactly as before:

File	What the dashboard puts there
extensions.conf	One line at the end: #tryinclude dvs_phone_extensions.conf
dvs_phone_extensions.conf	The whole Phone tab dialplan. Rewritten on every Save Phone; don't edit by hand.
rpt.conf	One private node per phone network (1000-1999), between the ; >>> dvs-phone and ; <<< dvs-phone marker lines.
dvs_phone_pjsip.conf / dvs_phone_iax*.conf	The SIP and IAX2 accounts, linked from pjsip.conf / iax.conf the same way.
/var/lib/asterisk/dvs_phone_*	Three tiny scripts: hang up, radio tone codes (*980-*982) and radio dial requests (*61).

Each phone network gets its own node

Instead of putting the patch on your radio node, each phone network (for example Hams Over IP on node 1001, VoIP.ms on node 1002) gets a private node with no radio of its own (rxchannel = Local/pseudo). While the Phone tab is open, the picked network's node is linked to your radio node. The node is set to simplex with voice keying so call audio passes cleanly to a simplex radio:

rpt.conf, written by the dashboard (Hams Over IP example)

```
[1001](node-main)
rxchannel = Local/pseudo
context = dvs-node-dvshoip          ; where its dialing rules start
functions = functions1001
phone_functions = functions1001
link_functions = functions1001
duplex = 1                          ; simplex, with the default VOX values below
voxtimeout = 10000
voxrecover = 2000
simplexpatchdelay = 25
simplexphonedelay = 25

[functions1001]
61 = autopatchup,noct=1,farenddisconnect=1,dialtime=20000,context=dvs-radio-dvshoip,quiet=1
62 = cmd,/var/lib/asterisk/dvs_phone_hangup
63 = cop,9                          ; phone patch on
64 = cop,10                         ; phone patch off
65 = cmd,/var/lib/asterisk/dvs_phone_hangup
99 = cop,6
980 = cmd,/var/lib/asterisk/dvs_phone_tonecode 0    ; sends *
981 = cmd,/var/lib/asterisk/dvs_phone_tonecode 1    ; sends #
982 = cmd,/var/lib/asterisk/dvs_phone_tonecode 2    ; sends *99
```

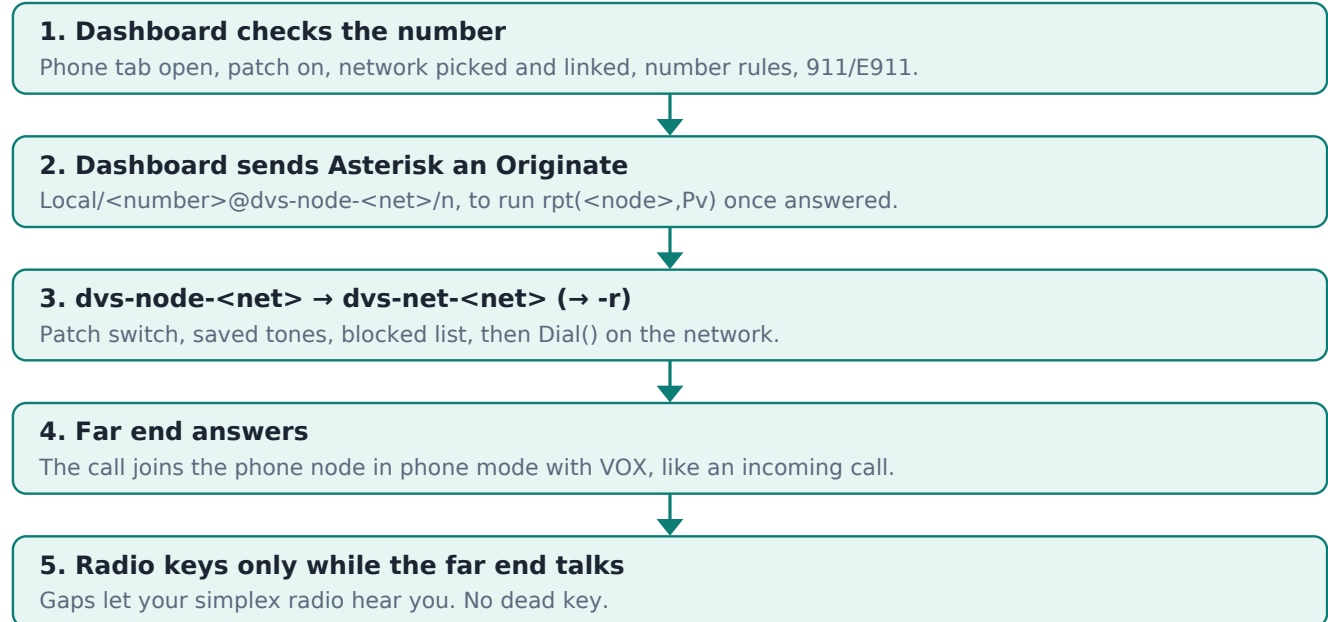
The contexts, one network at a time

Context	Job, in plain words
[dvs-radio-<net>]	Where *61 from the radio lands. It doesn't dial. It hands the number to the dashboard and hangs up, which ends the autopatch at once.

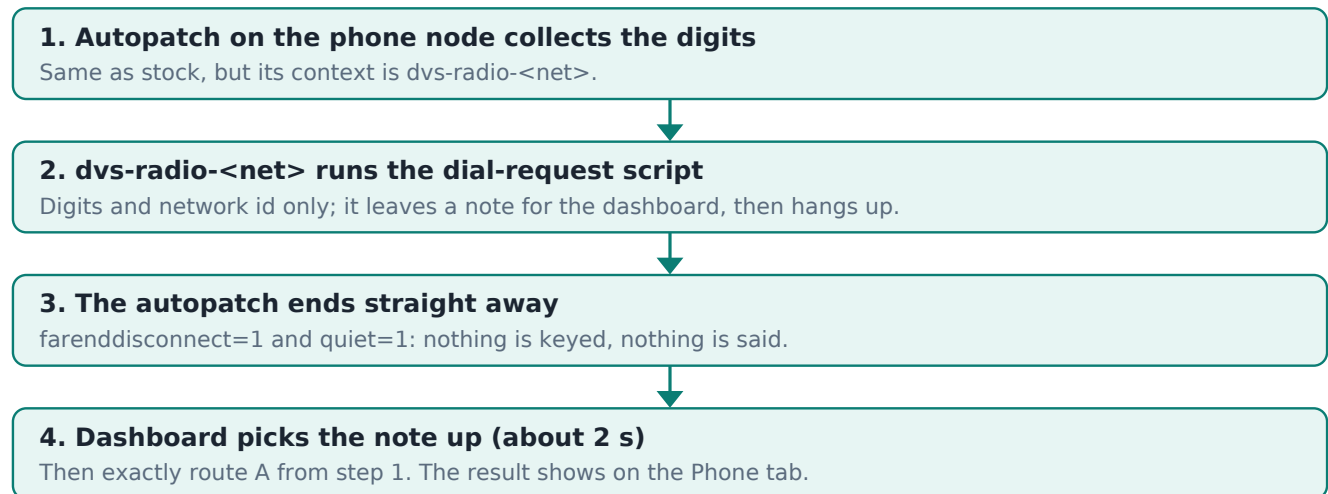
Context	Job, in plain words
[dvs-node-<net>]	Front door for outgoing calls . Checks the Phone patch on/off switch, picks up any saved favorite tones, then jumps to the network's rules.
[dvs-net-<net>]	The network's dialing rules : Test call (0098), Voicemail (0097), 911 only if E911 is ticked, number format, then Dial().
[dvs-net-<net>-r]	Phone-number networks only: the blocked list (toll-free, X00, X11, 555, 976, 809, 900), the same kinds stock check_route refuses.
[dvs-in-<net>]	Incoming calls . Busy unless the Phone tab is open with this network picked; then PIN, trusted numbers or straight through, and rpt(node,Pv).
Shared context	Job
[dvs-invalid]	Plays “no service” and hangs up (the stock invalidnum idea).
[dvs-radio-out] / [dvs-phone-out]	The same two jobs for the Phone Bridge node when no network owns it; they read the picked network from AstDB.
[dvs-tones]	Plays touch-tone recordings into a live call, for the “As sound” keypad setting.

How each kind of call flows

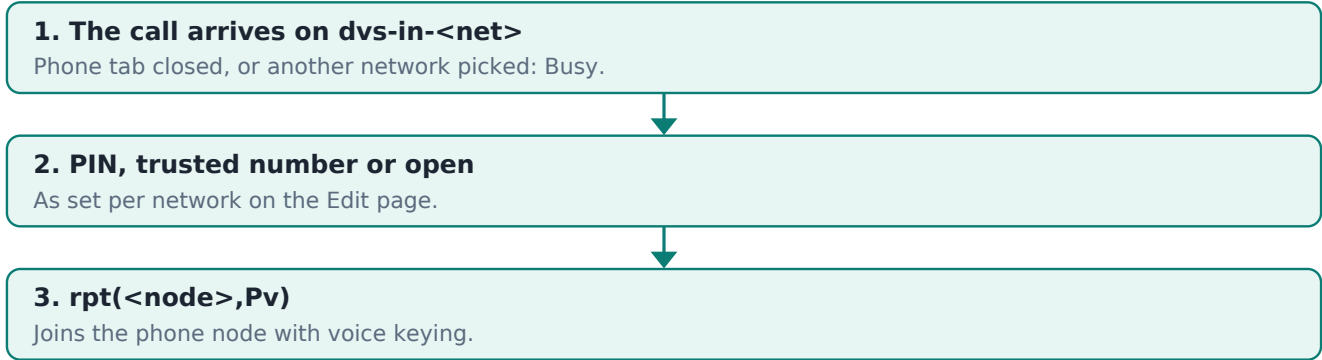
A. You press Dial (or a favorite) on the Phone tab



B. You key *61 and a number on the radio



C. Someone calls you



Hanging up and the other radio codes

Code	What it does now
*62 / *65	Hang up any phone call (both run the hang-up script). Before 9.3.71.10, *65 was autopatchdn and did nothing for these calls.
*63 / *64	Phone patch on / off (the dialplan's patch switch blocks new outgoing calls).
*980 *981 *982	Send *, # or *99 into the live call (the dashboard plays the tones).
*99	Stock phone-mode PTT code, kept as shipped.
Hang up button	Hangs up the call channels and cancels a radio dial that's still collecting digits (rpt cmd <node> autopatchdn).

Part 3 — Side by side

	Stock ASL3	ASL-DVS Pi02w
Phone calling out of the box	Off: codes commented out, pstn-out says “no service”	Off until you Activate it on the Edit page
Where the patch lives	On your radio node	On a private node per phone network, linked to the radio node only while the Phone tab is open
Files touched	You edit rpt.conf and extensions.conf by hand	Its own included files, plus marked blocks and one include line
Dialing from the radio	*61 autopatch dials directly	*61 hands the number to the dashboard, which places the call
Dialing from the web page	Not available	Phone tab Dial, favorites, Test call, Voicemail
How a call joins the node	Autopatch (keyed for the whole call)	rpt(node,Pv): phone mode with voice keying
Simplex radio while talking	Dead key: can't hear you	Keys only while the far end talks
Blocked numbers	check_route list	The same kinds, per network (dvs-net--r)
911	Routed like any X11 if you enable it	Only on a network with “E911 is set up” ticked
Incoming calls	Your own contexts (examples: iax-client)	dvs-in-: only while the Phone tab is open; PIN or trusted numbers
Listen-only option	No	Yes: monitor link, the radio never transmits into calls
Ringing heard on the radio	Yes (early audio passes through)	No: the radio hears the call once it's answered

Why the change: keeping stock autopatch would leave every simplex hotspot keyed for the whole call. Joining calls with rpt(node,Pv) uses app_rpt's own phone mode, the same path stock [iax-client] uses for IAX apps, so the radio follows the far end's voice instead.

Trade-offs to know: no ringback on the air; after 10 s of nonstop far-end talk the radio unkeys for 2 s (voxtimeout/voxtrecover) so you can break in; and radio-dialed calls need the Phone tab open, the only time your radio is linked to the phone node anyway.

Part 4 — Reading the generated file

These are real excerpts the dashboard writes for one Hams Over IP network (extension dialing) and one VoIP.ms network (phone numbers, PIN on incoming). Numbers are examples.

Outgoing front door and the *61 hand-off

```
[dvs-node-dvshoip]
exten => _X.,1,Set(DVSNET=dvshoip) ; which network this is
same => n,Set(DVSOPT=)
same => n,GotoIf($[${DB_EXISTS(dvsphone/then)}]?then:net) ; saved favorite tones waiting?
same => n(then),Set(DVSOPT=D(${DB_DELETE(dvsphone/then)})) ; send them after answer
same => n(net),GotoIf($[${DB(dvsphone/patch)}] = "0"?bad); patch switched off?
same => n,Goto(dvs-net-dvshoip,${EXTEN},1) ; on to the dialing rules
same => n(bad),Goto(dvs-invalid,s,1)

[dvs-radio-dvshoip]
exten => _X.,1,NoOp(radio dial)
same => n,System(/var/lib/asterisk/dvs_phone_dialreq "${EXTEN}" dvshoip)
same => n,Hangup() ; ends the autopatch
```

Phone-number network: rules and blocked list

```
[dvs-net-dvsvoipms]
exten => 911,1,NoOp(911 - E911 is set up on this network)
same => n,Set(CALLERID(num)=5550001111)
same => n,Dial(PJSIP/911@dvsvoipms,60)
exten => _NXXNXXXXXX,1,Goto(dvs-net-dvsvoipms-r,1${EXTEN},1) ; 10 digits: add the 1
exten => _1NXXNXXXXXX,1,Goto(dvs-net-dvsvoipms-r,${EXTEN},1)
exten => _X.,1,Goto(dvs-invalid,s,1) ; anything else: no service

[dvs-net-dvsvoipms-r]
exten => _1800NXXXXXX,1,Goto(dvs-invalid,s,1) ; blocked kinds ...
exten => _1NXXNXXXXXX,1,NoOp(dialing)
same => n,Set(CALLERID(num)=5550001111)
same => n,Dial(PJSIP/${EXTEN}@dvsvoipms,60,${DVSOPT})
```

Incoming calls

```
[dvs-in-dvsvoipms]
exten => _[+0-9].,1,NoOp(incoming call from ${CALLERID(num)})
same => n,GotoIf($[${DB(dvsphone/open)}] = "1" & "${DB(dvsphone/active)}" = "dvsvoipms"]?open)
same => n,Busy(20) ; tab closed or another network picked
same => n(open),NoOp()
same => n,GotoIf($[${CALLERID(num):-10}] = "5551234567"?connect) ; trusted number
same => n(ask),Read(PIN,beep,8,,1,10) ; ask for the PIN (3 tries)
...
same => n(connect),Answer()
same => n,Playback(rpt/connected)
same => n,rpt(1002,Pv) ; join node 1002 with voice keying
```

The AstDB switches the dialplan reads

Key	Set by the dashboard when...
dvsphone/open	The Phone tab is open (incoming calls are answered only then).
dvsphone/active	You pick a network (incoming calls for other networks get Busy).
dvsphone/patch	Phone patch on/off (0 blocks outgoing calls in dvs-node).
dvsphone/then	A favorite with saved tones is dialed; Dial() sends them after answer.

Checking it on the Pi

Command	Shows
<code>sudo asterisk -rx "dialplan show dvs-net-dvshoip"</code>	The loaded dialing rules for one network.
<code>sudo asterisk -rx "database show dvsphone"</code>	The four switches above.
<code>sudo asterisk -rx "rpt xnode 652702" grep -E "ALINKS TXKEYED"</code>	Link mode (T two-way, R listen only) and whether the radio is keyed.
<code>sudo asterisk -rx "core show channels concise"</code>	Live calls (PJSIP/dvs... or IAX2/dvs... lines).
SysMon → Phone → Dialing rules	The same rules, read from the files, with the show/hide note on the dvs-radio step.